

PYTHON LOOP ASSIGNMENT

Build Your Own Rule-Based Python Chatbot

Name	GARLAPATI HARSHITA
Register No.	99230040303
Subject	Python Programming
Assignment Type	30-Minute Student Task

1. Aim

To develop a simple rule-based conversational chatbot in Python using a continuous while True loop, conditional statements, string processing, and user input.

2. Objective

The chatbot should accept messages from the user, recognize predefined greetings and topics, generate appropriate responses, handle unknown inputs, and terminate when the user enters bye or exit.

3. Key Concepts Used

Concept	Purpose
input()	Accepts the user's name and chat messages.
.lower().strip()	Normalizes input for easier and case-insensitive matching.
while True	Keeps the chatbot conversation running continuously.
if / elif / else	Applies rules to decide the chatbot's response.
in operator	Recognizes flexible variations such as 'hello bot' or 'hey hi'.
break	Terminates the loop when the user exits.

4. Program

```
print("=" * 50)
print(" RULE-BASED PYTHON CHATBOT")
print("=" * 50)

name = input("What is your name? ").strip()
print("\nHello", name + "!")
print("You can talk to me.")
print("Type 'bye' or 'exit' to end the conversation.\n")

while True:
    message = input("You: ").lower().strip()

    # Exit condition
    if message == "bye" or message == "exit":
        print("Bot: Goodbye", name + "! Take care.")
        break

    # Greetings
    elif "hello" in message or "hi" in message:
        print("Bot: Hi", name + "! How can I help you today?")
```

```

# Topic 1: Food
elif "food" in message or "favorite food" in message:
    print("Bot: I really like biryani, but I'm curious what you enjoy eating!")

# Topic 2: Weather
elif "weather" in message:
    print("Bot: I can't check live weather, but I hope it's pleasant where you are!")

# Topic 3: College / Studies
elif "college" in message or "studies" in message or "study" in message:
    print("Bot: Studying Python is a great step toward building real projects!")

# Unknown message
else:
    print("Bot: Sorry, I don't understand that yet. "
          "Try asking me about food, weather, or college.")

```

5. Working / Logic

INPUT → PROCESS → RESPOND → REPEAT → EXIT

- The program first asks the user for their name.
- It enters a while True loop and waits for a message.
- The message is converted to lowercase and whitespace is removed using `.lower().strip()`.
- Conditional rules check for exit commands, greetings, and the three supported topics: food, weather, and college/studies.
- The chatbot prints an appropriate response and returns to the input step.
- When bye or exit is entered, break terminates the loop and a personalized goodbye is shown.

6. Requirement Checklist

- Ask the user's name before the main conversation.
- Use while True to keep the conversation running.
- Use `input()` to receive messages.
- Clean input using `.lower().strip()`.
- Recognize greetings such as hello and hi.
- Answer at least three different questions/topics (food, weather, college).
- Use if / elif / else for decision making.
- Use break to exit on bye or exit.
- Use the user's name in responses.
- Provide a useful response for unknown messages.
- Use the in operator for flexible matching.

7. Sample Output

```

=====
RULE-BASED PYTHON CHATBOT
=====
What is your name? Harshita

Hello Harshita!

```

You can talk to me.
Type 'bye' or 'exit' to end the conversation.

You: HELLO
Bot: Hi Harshita! How can I help you today?
You: what is your favorite food
Bot: I really like biryani, but I'm curious what you enjoy eating!
You: tell me about the weather
Bot: I can't check live weather, but I hope it's pleasant where you are!
You: how is college going
Bot: Studying Python is a great step toward building real projects!
You: what is your favorite movie
Bot: Sorry, I don't understand that yet. Try asking me about food, weather, or college.
You: bye
Bot: Goodbye Harshita! Take care.

8. Test Cases

Input	Expected Behavior
hello	Displays a greeting.
HELLO	Still displays a greeting after normalization.
hi	Displays a greeting.
hey hi there	Triggers greeting using the in operator.
favorite food	Displays the food-related response.
weather	Displays the weather-related response.
college / studies	Displays the college/studies response.
Unknown message	Displays the fallback response.
bye / exit	Displays goodbye message and terminates.

9. Conclusion

The rule-based chatbot demonstrates the use of Python loops, conditional statements, string methods, input handling, the in operator, and loop control using break. The program can be extended by adding more rules, topics, and responses.

Academic Note: This program was written independently as a student exercise. Each statement can be explained during live demonstration.